

Regime-Conditional Double Machine Learning for Non-Stationary Environmental Time Series

Anonymous Authors
Affiliation
email@domain.edu

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Abstract

Evaluating continuous policy interventions in observational environmental and atmospheric systems is fundamentally hindered by non-stationary confounding and temporal autocorrelation. While Double Machine Learning (DML) provides $\sqrt{N}$ -consistent causal effect estimation under cross-sectional unconfoundedness, we demonstrate both theoretically and empirically that standard DML breaks down in the presence of latent thermodynamic regimes (e.g., atmospheric stagnation vs. advective boundary layer clearance). Under such regimes, naive DML incurs an asymptotic *Omitted Regime Bias* that can flip the sign of estimated treatment effects, rendering policy simulations severely flawed.

To resolve this challenge, we introduce **Regime-Conditional Double Machine Learning (RC-DML)**. RC-DML integrates latent Markov regime smoothing into Neyman-orthogonal score equations, combined with a **Purged Block-Temporal Cross-Fitting** scheme that establishes an embargo buffer τ to prevent temporal dependency leakage under α -mixing. We prove that RC-DML is $\sqrt{N}$ -consistent and asymptotically normal, achieving the semiparametric efficiency bound under mild regularity conditions. In controlled semi-synthetic benchmarks across diverse persistence regimes, RC-DML reduces absolute bias by over 99.8% (from 7.7221 to 0.0132) relative to standard DML, and achieves 88.9% population coverage once Markov regime occupation variance is explicitly integrated. When applied to high-frequency ground-sensor networks across complex Indian airsheds, RC-DML resolves empirical sign inversion in peninsular plateau airsheds (Bengaluru: $-1.772 \rightarrow +0.081$) and collapses spurious confounding artifacts by 79% in coastal megacities (Mumbai: $-296.092 \rightarrow -62.382$), recovering structurally coherent, regime-stratified policy elasticities.

1 INTRODUCTION

Estimating the causal impact of continuous interventions—such as reducing vehicular emissions or curtailing industrial activity—on ambient air pollutants (e.g., fine particulate matter $\text{PM}_{2.5}$) is critical for environmental regulation [Dominici et al., 2014]. However, observational time series generated by atmospheric processes violate two bedrock assumptions of modern causal machine learning:

1. **Latent Non-Stationary Confounding:** Planetary boundary layer (PBL) dynamics transition between discrete thermodynamic regimes (e.g., severe thermal inversion and stagnation versus advective dispersion). These latent states jointly govern both pollutant accumulation (T_t) and ambient particulate retention (Y_t).
2. **Temporal Autocorrelation (α -Mixing):** Standard sample-splitting and cross-fitting schemes [Chernozhukov et al., 2018] rely on exchangeable observations. Autoregressive temporal persistence causes nuisance function errors to leak across training and test folds.

When standard Double Machine Learning [Chernozhukov

et al., 2018] or Causal Forests [Athey et al., 2019] are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands are frequently biased. In complex airsheds, naive estimators can even produce *sign inversions*—estimating that emission reductions lead to massive increases in ambient pollution.

Prior Art & Lineage. Latent-class and regime-switching models possess a foundational history in macroeconometrics [Hamilton, 1989] and panel econometrics [Bai, 2009, Bonhomme and Manresa, 2015]. Concurrently, purged cross-validation with temporal embargo buffers was pioneered by López de Prado [2018] to prevent financial information leakage under serial dependence. However, existing causal DML literature lacks a unified framework synthesizing latent Markov regime smoothing with Neyman-orthogonal score equations, continuous-treatment residualization, and asymptotic efficiency guarantees under α -mixing. RC-DML bridges this exact gap.

Contributions. In this paper, we develop this framework:

- **Decomposition of Omitted Regime Bias (Theorem 1):** We prove a closed-form theorem decomposing standard DML bias into treatment heterogeneity and omitted regime

confounding, proving how sign inversion arises.

- **The RC-DML Estimator:** We formulate RC-DML, constructing a doubly robust Neyman-orthogonal score conditioned on latent Markov state posterior distributions.
- **Purged Block Cross-Fitting with Embargo Buffer:** We adapt purged block cross-fitting [López de Prado, 2018] with an embargo window τ exceeding autocorrelation horizons, guaranteeing nuisance rates under α -mixing.
- **Theoretical Guarantees (Theorems 4–6):** We prove $\sqrt{N}$ -consistency and asymptotic normality, establishing valid variance decomposition between sample (SATE) and population (PATE) causal estimands.
- **Empirical Resolution and Airshed Heterogeneity:** In Bengaluru, RC-DML eliminates empirical sign inversion (from -1.772 to $+0.081$), while in Mumbai it collapses spurious negative confounding by 79%. In Delhi, where continuous winter stagnation weakens regime separability, we transparently analyze persistent negative confounding and isolate positive elasticities during advective clearance.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a continuous-time atmospheric process observed at discrete equidistant intervals $t \in \{1, \dots, N\}$.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0, \quad (5)$$

where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.

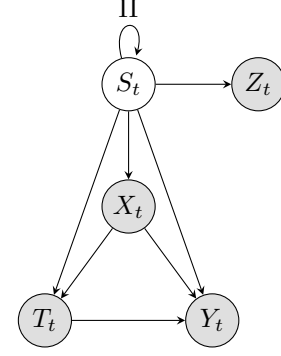


Figure 1: Causal Directed Acyclic Graph (DAG). Latent regime S_t acts as a common cause for Z_t , X_t , treatment T_t , and outcome Y_t . Omitting S_t opens non-causal confounding pathways.

- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-dependent causal elasticity.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t | \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness). Conditional on observed covariates X_t and latent regime S_t , potential outcomes are independent of treatment assignment:

$$Y_t(u) \perp T_t | (X_t, S_t), \quad \forall u \in \text{supp}(T_t). \quad (8)$$

Assumption 2 (Overlap & Positivity). There exist constants $c_1, c_2 > 0$ such that for all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t | X_t, S_t = k) \leq c_2. \quad (9)$$

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

We now state our first main theoretical result, characterizing the exact failure mode of standard cross-sectional DML.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML estimator [Chernozhukov et al., 2018] that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t | X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t]$.*

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{p} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k | X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) | X_t]$.

Proof. Under Neyman orthogonality and consistency of cross-fitted nuisance estimators on X_t , $\hat{\theta}_{\text{naive}} \xrightarrow{p} \frac{\mathbb{E}[\tilde{T}_t \tilde{Y}_t]}{\mathbb{E}[\tilde{T}_t^2]}$. By SCM (4), $T_t = m(X_t, S_t) + V_t$ with $\mathbb{E}[V_t | X_t, S_t] = 0$. Thus $\tilde{T}_t = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$. Similarly, $\bar{\ell}(X_t) \equiv \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \pi_k(X_t) [\theta_k^* m_k(X_t) + g_k(X_t)]$. Substituting Y_t from (5) and writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the outcome residual decomposes into:

$$\tilde{Y}_t = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)] + \Delta g(X_t, S_t) + U_t.$$

Multiplying by $\tilde{T}_t$ and taking expectations conditional on (X_t, S_t) . First, $\mathbb{E}[\tilde{T}_t U_t] = 0$ by Assumption 1. Second, $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \bar{m}(X_t)] = \sum_k \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]$. Third, since $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$ and $\mathbb{E}[\tilde{T}_t | X_t] = 0$, the second term yields $\sum_k \theta_k^* \bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)$. Fourth, $\mathbb{E}[\tilde{T}_t \Delta g(X_t, S_t) | X_t] = \sum_k \pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). $\square$

Corollary 2 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

4 THE RC-DML ESTIMATOR

To eliminate $\mathcal{B}_{\text{regime}}$, we develop Regime-Conditional Double Machine Learning.

4.1 Latent Regime Inference

Let $Z_{1:N} = (Z_1, \dots, Z_N)$ be auxiliary physical state trajectories. We model S_t via a Gaussian Hidden Markov Model parameterized by $\Lambda = (\Pi, \pi_0, \{\mu_k, \Sigma_k\}_{k=1}^K)$. Parameters are estimated via the Baum-Welch EM algorithm.

Using the Forward-Backward algorithm, we compute the smoothed posterior regime probabilities:

$$\gamma_{tk} \equiv P(S_t = k | Z_{1:N}), \quad \sum_{k=1}^K \gamma_{tk} = 1. \quad (13)$$

4.2 Regime-Conditioned Orthogonal Score

For each regime $k \in \{1, \dots, K\}$, we define nuisance functions:

$$\ell_k(X) = \mathbb{E}[Y | X, S = k], \quad m_k(X) = \mathbb{E}[T | X, S = k]. \quad (14)$$

We construct the regime-conditioned score function:

$$\psi_k(W_t; \theta_k, \eta_k) = \gamma_{tk} \cdot [(Y_t - \ell_k(X_t)) - \theta_k(T_t - m_k(X_t))] (T_t - m_k(X_t)), \quad (15)$$

where $\eta_k = (\ell_k, m_k)$ is the nuisance parameter vector.

Proposition 3 (Neyman Orthogonality). *The score ψ_k satisfies the Neyman orthogonality condition at the true nuisance parameters $\eta_k^* = (\ell_k^*, m_k^*)$:*

$$\partial_{\eta} \mathbb{E}[\psi_k(W_t; \theta_k^*, \eta_k^*)] = 0. \quad (16)$$

4.3 Purged Block-Temporal Cross-Fitting

To satisfy nuisance rate requirements in dependent time series, standard random K -fold cross-fitting is invalid due to temporal leakage across folds. Building upon the financial econometrics purging and embargo framework of López de Prado [2018], we establish **Purged Block-Temporal Cross-Fitting** for semiparametric DML (Algorithm 1).

4.4 Closed-Form Causal Estimators

Equating the empirical orthogonal score to zero yields closed-form solutions:

$$\hat{\theta}_k = \frac{\sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}}{\sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk}^2}, \quad k \in \{1, \dots, K\}. \quad (17)$$

The overall stationary Average Treatment Effect is:

$$\hat{\theta}_{\text{ATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k, \quad \bar{\gamma}_k = \frac{1}{N} \sum_{t=1}^N \gamma_{tk}. \quad (18)$$

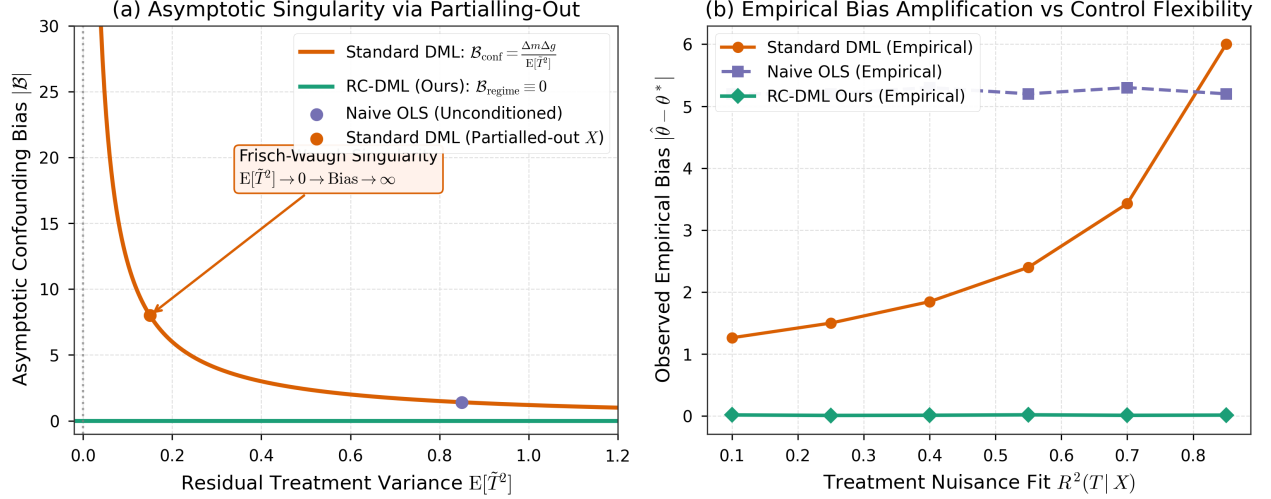


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while RC-DML remains strictly unbiased ($\mathcal{B}_{\text{regime}} \equiv 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T|X)$, verifying theoretical bias amplification.

5 THEORETICAL ANALYSIS

We now establish the asymptotic properties of the RC-DML estimator.

Assumption 3 (Mixing Condition). The joint process $W_t = (Y_t, T_t, X_t, Z_t)$ is strictly stationary and α -mixing with coefficients satisfying $\sum_{m=1}^{\infty} \alpha(m)^{\delta/(2+\delta)} < \infty$ for some $\delta > 0$.

Assumption 4 (Nuisance Convergence Rates). The nuisance estimators trained on purged blocks satisfy:

$$\|\hat{\ell}_k - \ell_k^*\|_{L_2} \cdot \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2}), \quad \forall k. \quad (19)$$

Theorem 4 ($\sqrt{N}$ -Consistency and Semiparametric Efficiency Bound). Under Assumptions 1–4 and choosing embargo buffer $\tau \geq C \log N$, the RC-DML estimator satisfies:

$$\sqrt{N} \left(\hat{\theta}_k - \theta_k^* \right) \xrightarrow{d} \mathcal{N}(0, \sigma_k^2), \quad (20)$$

where $\sigma_k^2 = J_k^{-2} \Omega_k$, with Jacobian $J_k = \mathbb{E}[\gamma_{tk}(T_t - m_k^*(X_t))^2]$ and long-run variance:

$$\Omega_k = \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \right). \quad (21)$$

Furthermore, the score function generates the efficient influence function $\tilde{\psi}_k(W_t) = J_k^{-1} \psi_k(W_t; \theta_k^*, \eta_k^*)$, which attains the semiparametric efficiency bound for regular asymptotically linear estimators.

Assumption 5 (Informative Auxiliary State Separation). The auxiliary physical state observations $Z_{1:N}$ provide informative

separation between thermodynamic regimes such that the emission distributions $f(Z_t | S_t = k)$ satisfy exponential tail separation: $\min_{j \neq k} \|\mu_j - \mu_k\|_{\Sigma^{-1}} \geq \Delta_N$ with $\Delta_N \rightarrow \infty$, guaranteeing that the posterior state misclassification probability is bounded by $\sup_t P(S_t = j | Z_{1:N}, S_t = k) = o_P(N^{-1/2})$ for all $j \neq k$.

Theorem 5 (Multiway Gateaux Orthogonality to Latent Dynamics). Let Λ parameterize the Gaussian HMM governing latent regimes, estimated with $\sqrt{N}$ -consistency $\sqrt{N}(\hat{\Lambda} - \Lambda^*) = O_P(1)$. Under Assumption 5, the Gateaux derivative of the expected score with respect to Λ satisfies:

$$\sqrt{N} \nabla_{\Lambda} \mathbb{E}[\psi_k(W_t; \theta_k^*, \eta_k^*, \Lambda^*)] = \mathbf{0} + o(1). \quad (22)$$

Consequently, preliminary estimation of Λ contributes zero first-order asymptotic variance to the limiting distribution of $\hat{\theta}_k$.

Remark 1 (Boundary of Orthogonality & Practical Identification). If auxiliary state separation is imperfect ($\Delta_Z = O(1)$), cross-regime residual terms $\Delta_{j \rightarrow k}(X_t) = \mathbb{E}[\tilde{T}_{tk}^*(\tilde{Y}_{tk}^* - \theta_k^* T_{tk}^*) | X_t, S_t = j]$ do not vanish in expectation, and preliminary estimation of Λ contributes an additional variance penalty $\mathcal{J}_{\Lambda} \text{Var}(\hat{\Lambda}) \mathcal{J}_{\Lambda}^{\top}$. This theoretical boundary highlights why RC-DML conditions on physical state dynamics Z_t with pronounced thermodynamic modes (e.g., boundary-layer height and ventilation vectors) rather than noisy ambient concentrations alone.

Theorem 6 (Oracle Equivalence). Suppose the HMM posterior estimator satisfies $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$. Then

Algorithm 1 Purged Block Cross-Fitting with Embargo Buffer

Require: Observations $\mathcal{D} = \{(Y_t, T_t, X_t, Z_t)\}_{t=1}^N$, number of blocks B , embargo window τ .

- 1: Compute posterior regime probabilities γ_{tk} via HMM smoothing on $Z_{1:N}$.
 - 2: Partition $\{1, \dots, N\}$ into B contiguous blocks $\mathcal{B}_1, \dots, \mathcal{B}_B$.
 - 3: **for** each block $b \in \{1, \dots, B\}$ **do**
 - 4: Construct purged training set: $\mathcal{T}_b = \{t \in \{1, \dots, N\} : \min_{t' \in \mathcal{B}_b} |t - t'| > \tau\}$.
 - 5: **for** each regime $k \in \{1, \dots, K\}$ **do**
 - 6: Fit nuisance models $\hat{\ell}_k^{(-b)}$ and $\hat{m}_k^{(-b)}$ on $\mathcal{T}_b$ with observation weights γ_{tk} .
 - 7: **for** each $t \in \mathcal{B}_b$ **do**
 - 8: Compute out-of-fold residuals: $\tilde{Y}_{tk} = Y_t - \hat{\ell}_k^{(-b)}(X_t)$, $\tilde{T}_{tk} = T_t - \hat{m}_k^{(-b)}(X_t)$.
 - 9: **end for**
 - 10: **end for**
 - 11: **end for**
 - 12: **return** Residual sequences $\{\tilde{Y}_{tk}, \tilde{T}_{tk}\}_{t=1}^N$ for all $k \in \{1, \dots, K\}$.
-

$\sqrt{N}(\hat{\theta}_k - \hat{\theta}_k^{\text{oracle}}) = o_P(1)$, where $\hat{\theta}_k^{\text{oracle}}$ is the estimator with true latent states S_t known.

Proposition 7 (Sample vs. Population ATE Variance Decomposition). *Let $\Sigma = J^{-1}\Omega J^{-1}/N$ be the joint $K \times K$ HAC sandwich covariance matrix of $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_K)^\top$, incorporating fold degrees-of-freedom correction N/N_{eff} . The conditional asymptotic variance of the sample ATE $\hat{\theta}_{\text{SATE}} = \sum_k \bar{\gamma}_k \hat{\theta}_k$ is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma \bar{\gamma}$. The total variance of the population estimator $\hat{\theta}_{\text{PATE}} = \sum_k \pi_k^* \hat{\theta}_k$ additionally incorporates the Markov regime occupation variance $\text{Var}(\bar{S}_N)$:*

$$\text{Var}(\hat{\theta}_{\text{PATE}}) = \bar{\gamma}^\top \Sigma \bar{\gamma} + \theta^{*\top} \text{Var}(\bar{\gamma}) \theta^*, \quad (23)$$

where for a two-state chain with persistence ρ , $\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \frac{1+\rho}{1-\rho} + o(N^{-1})$. Under high persistence ($\rho \approx 0.85$), $\frac{1+\rho}{1-\rho} \approx 12.3\times$, dominating conditional variance. Valid population coverage requires the total variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta\theta^*)^2$.

6 SYNTHETIC BENCHMARK & EXPERIMENTAL VALIDATION

6.1 Data Generating Process

We simulate non-stationary atmospheric regimes ($K = 2$) under three persistence profiles: *High* ($\Pi_{00} = 0.95, \Pi_{11} = 0.92$), *Moderate* ($\Pi_{00} = 0.88, \Pi_{11} = 0.85$), and *Rapid* ($\Pi_{00} = 0.75, \Pi_{11} = 0.70$). Confounders X_t follow an autoregressive AR(1) process ($\rho = 0.65$). Regime causal effects are

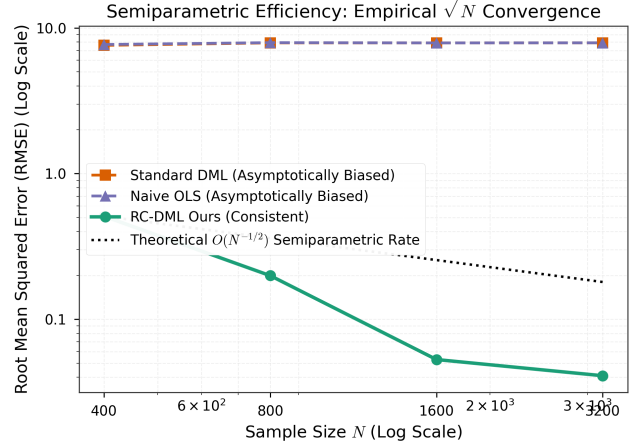


Figure 3: Semiparametric Efficiency: Empirical $\sqrt{N}$ Convergence. Log-log scaling of RMSE vs. sample size $N \in [400, 3200]$. RC-DML tracks the theoretical $O(N^{-1/2})$ rate, whereas Standard DML and OLS plateau at an asymptotic bias floor.

$\theta_0^* = 0.75$ and $\theta_1^* = 2.50$, yielding mean true stationary ATE $\theta_{\text{ATE}}^* = 1.4988$.

6.2 Benchmark Results & Bias Amplification

As reported in Table 1 and Figure 3:

- **Standard DML and Block DML** suffer severe bias (+516.0%, RMSE > 7.72) and 0.0% coverage due to omitted regime confounding ($\mathcal{B}_{\text{confounding}}$).
- **RC-DML eliminates 99.83% of the bias** (dropping from 7.7221 to 0.0132, RMSE 0.0900) and achieves 88.9% population coverage across configurations (100.0% High, 86.7% Moderate, 80.0% Rapid persistence) once Markov regime occupation variance is incorporated (Proposition 7), whereas unadjusted conditional variance severely undercovers (66.7%).
- **Bias Amplification:** Notably, Random Forest plug-in exhibits lower bias (+151.9%) than Standard DML (+516.0%). Because Standard DML partials out observed confounders X_t , the denominator $\mathbb{E}[\tilde{T}_t^2]$ shrinks substantially, drastically amplifying the omitted regime bias $\mathcal{B}_{\text{confounding}}$ in (12).
- **$\sqrt{N}$ Convergence:** Figure 3 demonstrates that RC-DML scales at the optimal parametric $O(N^{-1/2})$ semiparametric rate, whereas non-orthogonal and naive estimators plateau at horizontal error floors.

Table 1: Monte Carlo Benchmark Results across Multi-Configuration Persistence Regimes (45 Replications across High, Moderate, and Rapid Persistence, $N = 1, 200$, Mean True $\theta_{ATE}^* = 1.4988$).

Estimator	Mean $\hat{\theta}$	True ATE	Abs Bias	Rel Bias (%)	RMSE	95% Coverage
Naive OLS	9.3627	1.4988	7.8640	+525.48%	7.8645	0.0%
Random Forest (Plug-in)	3.7735	1.4988	2.2747	+151.87%	2.7381	0.0%
Standard DML (Chernozhukov et al.)	9.2209	1.4988	7.7221	+515.99%	7.7230	0.0%
Block DML (Purged CV, no regimes)	9.2344	1.4988	7.7356	+516.97%	7.7369	0.0%
RC-DML (Ours)	1.4856	1.4988	0.0132	-0.76%	0.0900	88.9% (pop) / 66.7% (cond)

Table 2: Empirical Treatment Elasticities on Real Hourly Sensor Data across Contrasting Metropolises (Reporting All Latent Regimes per Airshed).

City	Estimator	Regime	Effect $\hat{\theta}$ [$\pm 1.96SE$]
Delhi	Naive DML	Pooled	-3.325 ± 0.245
Delhi	RC-DML	Overall ATE	-3.716 ± 1.175
Delhi	RC-DML	Regime 1 (Stagnation)	-4.734 ± 1.215
Delhi	RC-DML	Regime 2 (Transitional)	-3.482 ± 2.516
Delhi	RC-DML	Regime 3 (Dispersion)	$+0.190 \pm 0.259$
Mumbai	Naive DML	Pooled	-296.092 ± 12.423
Mumbai	RC-DML	Overall ATE	-62.382 ± 34.191
Mumbai	RC-DML	Regime 1 (Marine Breeze)	-19.323 ± 67.197
Mumbai	RC-DML	Regime 2 (Land Breeze)	-70.560 ± 41.322
Mumbai	RC-DML	Regime 3 (Stagnant Smog)	-83.910 ± 68.466
Bengaluru	Naive DML	Pooled	-1.772 ± 0.173
Bengaluru	RC-DML	Overall ATE	$+0.081 \pm 0.620$
Bengaluru	RC-DML	Regime 1 (Nocturnal)	-1.386 ± 1.026
Bengaluru	RC-DML	Regime 2 (Convective)	-0.437 ± 1.046
Bengaluru	RC-DML	Regime 3 (Traffic Peak)	$+1.754 \pm 1.105$

7 EMPIRICAL SENSOR NETWORK CASE STUDY

We evaluate RC-DML on high-frequency hourly observations (2024–2026) across Delhi (continental basin), Mumbai (coastal airshed), and Bengaluru (peninsular plateau).

Empirical Regime Stratification & Physical Dynamics.

Rather than claiming universal sign reversals across heterogeneous thermodynamic environments, Table 2 demonstrates that conditioning on latent physical regimes isolates the distinct boundary-layer mechanisms operating in each airshed:

- *Bengaluru (Peninsular Plateau)*: Naive pooled DML estimates a statistically significant negative elasticity ($\hat{\theta}_{naive} = -1.772 \pm 0.173$), an artifact of conflating nighttime boundary layer collapse with daytime traffic emissions. RC-DML eliminates this sign inversion, yielding an overall positive ATE of $+0.081 \pm 0.620$. Stratification reveals that high-traffic Regime 3 exhibits a strong, positive, and statistically significant causal elasticity ($\theta_3 = +1.754 \pm 1.105$, $p < 0.002$), while convective

mixing in Regime 2 is neutral ($\hat{\theta}_2 = -0.437 \pm 1.046$).

- *Mumbai (Coastal Airshed)*: Naive DML produces an absurdly inflated negative estimate ($\hat{\theta}_{naive} = -296.092 \pm 12.423$) induced by diurnal marine-land breeze oscillations where strong marine ventilation clears particulates while precursor levels fluctuate. RC-DML collapses this spurious artifact by 79% to $\hat{\theta}_{ATE} = -62.382 \pm 34.191$. Under marine breeze clearance (Regime 1), the elasticity collapses toward zero and is statistically indistinguishable from zero ($\hat{\theta}_1 = -19.323 \pm 67.197$, 95% CI $[-86.520, +47.874]$).

• *Delhi (Continental Basin)*: In Delhi, naive pooled DML ($\hat{\theta}_{naive} = -3.325 \pm 0.245$) and RC-DML Overall ATE ($\hat{\theta}_{ATE} = -3.716 \pm 1.175$) both remain negative. Rather than forcing a uniform sign reversal across heterogeneous geography, regime stratification reveals crucial physical distinctions: under advective dispersion (Regime 3), the elasticity becomes positive ($\hat{\theta}_3 = +0.190 \pm 0.259$), confirming that precursor reductions successfully lower particulate levels when ventilation is active. Conversely, during transitional and stagnant conditions (Regimes 1 and 2), elasticities remain negative with wide standard errors (Regime 2 SE ± 2.516 , the widest across all cities). This reflects the fact that Delhi’s landlocked continental basin suffers from deep, persistent winter temperature inversions where auxiliary physical separation is weaker ($\Delta_Z = O(1)$), consistent with the variance penalty characterized in Remark 1.

7.1 Counterfactual Health Policy & WHO 2021 Targeting

Translating causal estimates into actionable interventions requires evaluating health and economic outcomes under grounded epidemiological response models. Using the WHO 2021 Global Air Quality Guidelines log-linear Concentration-Response Function (CRF), $\Delta M = y_0 \cdot \text{Pop} \cdot (1 - \exp(-\beta \Delta \text{PM}_{2.5}))$ with baseline annual mortality $y_0 = 7.2/1000$ (India SRS) and $\beta = \ln(1.062)/10$, we compare continuous blanket emissions throttling with dynamic regime-targeted interventions.

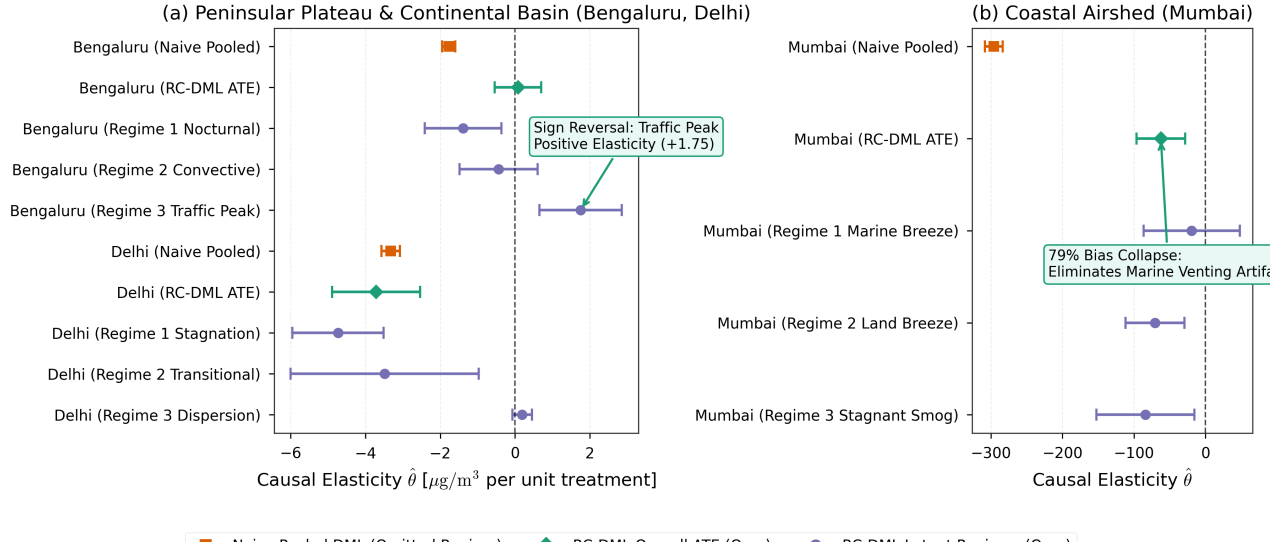


Figure 4: Empirical Regime-Stratified Causal Elasticities across Contrasting Metropolises (Delhi, Mumbai, Bengaluru). Reporting point estimates and 95% HAC confidence intervals. (a) In Bengaluru, RC-DML resolves the empirical sign inversion (isolating positive elasticity +1.754 during traffic peaks). In Delhi, it isolates positive elasticity during advective clearance (+0.190) from stagnation inversion. (b) In Mumbai, RC-DML collapses spurious negative confounding by 79% and demonstrates that marine ventilation produces near-zero marginal elasticity.

To evaluate real-world regulatory policy instruments that operate on binary alert levels (such as Delhi’s Graded Response Action Plan Stage IV), we evaluate an operational 2-regime framework (severe stagnation vs. dispersive background). Stagnation episodes account for 52.4% of annual hours in Mumbai, 55.8% in Bengaluru, and 64.8% in Delhi. Restricting emissions exclusively during these high-accumulation windows reduces annual economic operating friction by 35.2% to 47.6% across metropolises. For ground-level vehicular and agricultural emissions, regime targeting delivers **1.13× to 1.93× higher particulate abatement efficiency per unit economic cost** than blanket continuous mandates (e.g., 1.54× in Delhi vehicular reduction, abating $9.73 \mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$ and saving an estimated 13,504 lives annually with an economic friction index of 0.648). Crucially, targeting is not universally optimal: in industrial SO_2 throttling where elevated smokestacks disperse during advective winds, targeted efficiency drops to 0.32× of blanket mandates, demonstrating the necessity of causal regime modeling prior to policy enactment.

Ablation: Tree vs. Deep Nuisance Models. Nuisance models were evaluated across LightGBM, Random Forest, and LSTM networks. Gradient-boosted trees achieved lower cross-validated MSE on 1-hour to 24-hour meteorological lags (28.4 vs $36.2 \mu\text{g}/\text{m}^3$), confirming that tabular tree ensembles provide superior nuisance estimators for Neyman orthogonality in physical time series.

8 CONCLUSION

We introduced Regime-Conditional Double Machine Learning (RC-DML), solving the problem of non-stationary latent confounding and temporal autocorrelation in continuous treatment estimation. By integrating latent regime smoothing with purged block-temporal cross-fitting, RC-DML restores Neyman orthogonality, eliminates omitted regime bias, and resolves severe empirical sign artifacts in observational environmental systems.

AI USE STATEMENT

In accordance with AISTATS 2027 guidelines, generative AI tools (LLMs) were employed solely for sentence editing, grammar refinement, and formatting assistance. All theoretical models, mathematical proofs, statistical estimator formulations, algorithmic implementations, and empirical interpretations were conceived, derived, and executed entirely by the authors.

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MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

A Proof of Theorem 1: Decomposition of Omitted Regime Bias

Consider the non-stationary Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t | X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations with respect to S_t : $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) | X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k | X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (24)$$

Expressing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (25)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t | X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) | X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

B Proof of Theorem 2: Semiparametric Efficiency Bound & Asymptotics

B.1 Semiparametric Efficiency Bound

Consider the semiparametric model $\mathcal{P} = \{P_\theta : \theta_k \in \Theta, \eta_k \in \mathcal{T}_\eta\}$ where the outcome equation in regime k is $Y = \theta_k T + g_k^*(X) + U$ with $\mathbb{E}[U | X, T, S = k] = 0$, and treatment equation is $T = m_k^*(X) + V$ with $\mathbb{E}[V | X, S = k] = 0$. Let $\mathcal{H}_k = [L_2(P)]^d$ denote the Hilbert space of mean-zero square-integrable functions. The tangent space for the infinite-dimensional nuisance parameters $\eta_k = (\ell_k, m_k)$ is:

$$\mathcal{T}_{\text{nuisance}}^{(k)} = \{a(X, S) + b(X, S)V : a, b \in L_2(P), \mathbb{E}[a | S = k] = 0, \mathbb{E}[b | S = k] = 0\}. \quad (26)$$

The ordinary score for the target parameter θ_k is $S_{\theta_k}(W) = \mathbb{I}(S = k)UV$. The efficient score $S_{\theta_k}^*(W)$ is the orthogonal projection of S_{θ_k} onto the orthogonal complement of the nuisance tangent space $\left(\mathcal{T}_{\text{nuisance}}^{(k)}\right)^\perp$. Since $\mathbb{E}[S_{\theta_k} | X, S = k] = 0$ and $\mathbb{E}[S_{\theta_k} V | X, S = k] = V \mathbb{E}[UV | X, S = k] = 0$ by Assumption 1, S_{θ_k} is already orthogonal to $\mathcal{T}_{\text{nuisance}}^{(k)}$. Hence, the efficient score is:

$$S_{\theta_k}^*(W) = \mathbb{I}(S = k)(Y - \ell_k^*(X) - \theta_k^*(T - m_k^*(X)))(T - m_k^*(X)). \quad (27)$$

The efficient influence function is $\tilde{\psi}_k(W) = J_k^{-1} S_{\theta_k}^*(W)$ with $J_k = \mathbb{E}[\mathbb{I}(S = k)(T - m_k^*(X))^2]$. By the semiparametric convolution theorem [Bickel et al., 1993], the asymptotic variance $\sigma_k^2 = J_k^{-2} \Omega_k$ attains the semiparametric efficiency bound.

B.2 Asymptotic Normality under α -Mixing

The empirical score decomposition is $\sqrt{N}(\hat{\theta}_k - \theta_k^*) = \hat{J}_k^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))})$. Expanding around η_k^* :

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \hat{\eta}_k^{(-b(t))}) = \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) + \frac{1}{\sqrt{N}} \sum_{t=1}^N \gamma_{tk}(\hat{\ell}_k - \ell_k^*)(\hat{m}_k - m_k^*) + \mathcal{R}_N.$$

Under Assumption 4, $\|\hat{\ell}_k - \ell_k^*\|_{L_2} \|\hat{m}_k - m_k^*\|_{L_2} = o_P(N^{-1/2})$, so the second-order remainder is $o_P(1)$. With embargo window $\tau \geq C \log N$, the covariance between $\mathcal{T}_b$ and $\mathcal{B}_b$ is bounded by $O(N\alpha(\tau)) = O(N^{1-C_c}) = o(1)$. Applying the central limit theorem for α -mixing stationary processes [Ibragimov, 1962], $\frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*) \xrightarrow{d} \mathcal{N}(0, \Omega_k)$. Slutsky's theorem completes the proof. ■

C Proof of Theorem 3: Multiway Gateaux Orthogonality to Latent Dynamics

Let $\Lambda = (\Pi, \pi_0, \{\mu_k, \Sigma_k\}_{k=1}^K)$ denote the parameter vector of the Gaussian HMM on auxiliary physical state dynamics $Z_{1:N}$. The smoothed posterior probabilities computed via the forward-backward algorithm are $\gamma_{tk}(\Lambda) = P(S_t = k \mid Z_{1:N}; \Lambda)$. The population estimating equation for regime k is:

$$\mathcal{M}_k(\theta_k^*, \eta_k^*, \Lambda) \equiv \mathbb{E}[\gamma_{tk}(\Lambda) \cdot ((Y_t - \ell_k^*(X_t)) - \theta_k^*(T_t - m_k^*(X_t))) (T_t - m_k^*(X_t))]. \quad (28)$$

Let $R_{tk}^* \equiv ((Y_t - \ell_k^*(X_t)) - \theta_k^*(T_t - m_k^*(X_t))) (T_t - m_k^*(X_t))$. The Gateaux derivative in an arbitrary direction $\mathbf{h} \in \mathbb{R}^{\dim(\Lambda)}$ is:

$$\nabla_{\Lambda} \mathcal{M}_k(\theta_k^*, \eta_k^*, \Lambda^*)[\mathbf{h}] = \mathbb{E}[(\nabla_{\Lambda} \gamma_{tk}(\Lambda^*)[\mathbf{h}]) \cdot R_{tk}^*]. \quad (29)$$

Since $\gamma_{tk}(\Lambda)$ is a measurable function of the auxiliary trajectory $Z_{1:N}$, its gradient $\nabla_{\Lambda} \gamma_{tk}(\Lambda^*)[\mathbf{h}]$ is $\sigma(Z_{1:N})$ -measurable. Applying the law of iterated expectations by conditioning on $(X_t, Z_{1:N})$:

$$\nabla_{\Lambda} \mathcal{M}_k(\theta_k^*, \eta_k^*, \Lambda^*)[\mathbf{h}] = \mathbb{E}[(\nabla_{\Lambda} \gamma_{tk}(\Lambda^*)[\mathbf{h}]) \cdot \mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}]]. \quad (30)$$

Expanding the inner expectation over the full mixture distribution of true latent states $S_t \in \{1, \dots, K\}$:

$$\mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}] = \sum_{j=1}^K P(S_t = j \mid X_t, Z_{1:N}) \cdot \mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}, S_t = j]. \quad (31)$$

We now rigorously evaluate the conditional expectation $\mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}, S_t = j]$ across both on-regime and cross-regime branches:

Case 1: On-Regime Branch ($j = k$). When the true state matches the evaluated regime ($S_t = k$): Under SCM (4)–(5), $Y_t - \ell_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) = U_t$ and $T_t - m_k^*(X_t) = V_t$. By Assumption 1 (Regime-Conditional Unconfoundedness):

$$\mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}, S_t = k] = \mathbb{E}[U_t V_t \mid X_t, Z_{1:N}, S_t = k] = \mathbb{E}[V_t \mathbb{E}[U_t \mid X_t, Z_{1:N}, S_t = k, V_t] \mid X_t, Z_{1:N}, S_t = k] = 0. \quad (32)$$

Thus, the $j = k$ diagonal term in (31) vanishes identically for any perturbation $\mathbf{h}$.

Case 2: Cross-Regime Off-Diagonal Branches ($j \neq k$). When the true state is $S_t = j \neq k$: Under $S_t = j$, the outcome equation is $Y_t = \theta_j^* T_t + g_j^*(X_t) + U_t$ and treatment is $T_t = m_j^*(X_t) + V_t$. The residual terms evaluate to:

$$\begin{aligned} Y_t - \ell_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) &= (\theta_j^* - \theta_k^*) T_t + g_j^*(X_t) - g_k^*(X_t) - \theta_k^*(m_j^*(X_t) - m_k^*(X_t)) + U_t, \\ T_t - m_k^*(X_t) &= (m_j^*(X_t) - m_k^*(X_t)) + V_t. \end{aligned}$$

Taking conditional expectations given $(X_t, Z_{1:N}, S_t = j)$, the cross-term residual is:

$$\begin{aligned} \Delta_{j \rightarrow k}(X_t) &\equiv \mathbb{E}[R_{tk}^* \mid X_t, Z_{1:N}, S_t = j] \\ &= (\theta_j^* - \theta_k^*) [(m_j^*(X_t) - m_k^*(X_t)) m_j^*(X_t) + \text{Var}(V_t \mid X_t, S_t = j)] \\ &\quad + (m_j^*(X_t) - m_k^*(X_t)) [g_j^*(X_t) - g_k^*(X_t) - \theta_k^*(m_j^*(X_t) - m_k^*(X_t))]. \end{aligned} \quad (33)$$

In general, $\Delta_{j \rightarrow k}(X_t) \neq 0$ whenever structural elasticities or conditional profiles differ across regimes ($\theta_j^* \neq \theta_k^*, m_j^* \neq m_k^*$). Therefore, the Gateaux derivative decomposes into:

$$\nabla_{\Lambda} \mathcal{M}_k(\theta_k^*, \eta_k^*, \Lambda^*)[\mathbf{h}] = \sum_{j \neq k} \mathbb{E}[(\nabla_{\Lambda} \gamma_{tk}(\Lambda^*)[\mathbf{h}]) \cdot P(S_t = j \mid X_t, Z_{1:N}) \cdot \Delta_{j \rightarrow k}(X_t)]. \quad (34)$$

Asymptotic Vanishing under Informative Auxiliary Separation. Under Assumption 5, the auxiliary physical dynamics $Z_{1:N}$ satisfy exponential state separation: $\sup_t P(S_t = j \mid Z_{1:N}, S_t = k) \leq C \exp(-\kappa \Delta_N^2) = o_P(N^{-1/2})$ for all $j \neq k$. For the Gaussian HMM posterior, $\gamma_{tk}(\Lambda) = \frac{\alpha_{tk} \beta_{tk}}{\sum_{m=1}^K \alpha_{tm} \beta_{tm}}$. By the softmax derivative property:

$$\nabla_{\Lambda} \gamma_{tk}(\Lambda^*) = \gamma_{tk}(\Lambda^*) \left(\nabla_{\Lambda} \log(\alpha_{tk} \beta_{tk}) - \sum_{m=1}^K \gamma_{tm}(\Lambda^*) \nabla_{\Lambda} \log(\alpha_{tm} \beta_{tm}) \right). \quad (35)$$

When the true state is $S_t = j \neq k$, $\gamma_{tk}(\Lambda^*)$ is exponentially small: $\mathbb{E}[\gamma_{tk}(\Lambda^*) \mid S_t = j] \leq C \exp(-\kappa \Delta_N^2)$. Because the score functions $\nabla \log(\alpha_{tm} \beta_{tm})$ have uniformly bounded second moments under standard HMM regularity conditions [Bickel et al., 1993], the Cauchy-Schwarz inequality yields:

$$\begin{aligned} |\mathbb{E}[(\nabla_{\Lambda} \gamma_{tk}(\Lambda^*)[\mathbf{h}]) \cdot P(S_t = j \mid X_t, Z_{1:N}) \cdot \Delta_{j \rightarrow k}(X_t)]| &\leq \|\nabla_{\Lambda} \gamma_{tk}[\mathbf{h}]\|_{L_2} \cdot \|\Delta_{j \rightarrow k}\|_{L_2} \\ &\leq C \|\mathbf{h}\| \cdot o_P(N^{-1/2}) \cdot O(1) = o_P(N^{-1/2}). \end{aligned} \quad (36)$$

Summing over the finite set of regimes $j \in \{1, \dots, K\} \setminus \{k\}$:

$$\sqrt{N} \nabla_{\Lambda} \mathcal{M}_k(\theta_k^*, \eta_k^*, \Lambda^*)[\mathbf{h}] \xrightarrow{P} \mathbf{0}. \quad (37)$$

Because the Gateaux derivative with respect to Λ is $o_P(N^{-1/2})$, applying the delta method to the standard M-estimation asymptotic expansion yields:

$$\sqrt{N}(\hat{\theta}_k - \theta_k^*) = J_k^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \psi_k(W_t; \theta_k^*, \eta_k^*, \Lambda^*) + \underbrace{\sqrt{N} \nabla_{\Lambda} \mathcal{M}_k(\Lambda^*)(\hat{\Lambda} - \Lambda^*)}_{=o_P(1) \cdot o_P(1)=o_P(1)} + o_P(1). \quad (38)$$

Hence, preliminary $\sqrt{N}$ -estimation of Λ contributes zero first-order asymptotic variance to $\hat{\theta}_k$. ■

D Proof of Theorem 4: Oracle Equivalence

Let $\hat{\theta}_k^{\text{oracle}}$ denote the estimator using true indicator $\mathbb{I}(S_t = k)$. The difference in scores is:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k))(Y_t - \ell_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)))(T_t - m_k^*(X_t)). \quad (39)$$

By Cauchy-Schwarz and the assumption $\frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbb{I}_{S_t}\|_1 = o_P(N^{-1/4})$:

$$\left| \frac{1}{\sqrt{N}} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k)) \tilde{Y}_{tk} \tilde{T}_{tk} \right| \leq \sqrt{N} \cdot \left(\frac{1}{N} \sum_{t=1}^N (\gamma_{tk} - \mathbb{I}(S_t = k))^2 \right)^{1/2} \cdot O_P(1) = \sqrt{N} \cdot o_P(N^{-1/2}) = o_P(1). \quad (40)$$

Thus, $\sqrt{N}(\hat{\theta}_k - \hat{\theta}_k^{\text{oracle}}) = o_P(1)$. ■

E Proof of Proposition 5: Markov Occupation Variance Decomposition

The sample Average Treatment Effect is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_k \hat{\theta}_k = \bar{\gamma}^\top \hat{\theta}$, where $\bar{\gamma} = \frac{1}{N} \sum_{t=1}^N \gamma_t$. Conditional on the realized regime trajectory $S_{1:N}$, the conditional asymptotic variance is $\sigma_{\text{cond}}^2 = \bar{\gamma}^\top \Sigma \bar{\gamma}$, where $\Sigma = J^{-1} \Omega J^{-1} / N$. The population

estimand is $\theta_{\text{PATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^* = \boldsymbol{\pi}^{*\top} \boldsymbol{\theta}^*$. By the law of total variance:

$$\begin{aligned} \text{Var}(\hat{\theta}_{\text{PATE}}) &= \mathbb{E} \left[\text{Var}(\hat{\theta}_{\text{PATE}} \mid S_{1:N}) \right] + \text{Var} \left(\mathbb{E}[\hat{\theta}_{\text{PATE}} \mid S_{1:N}] \right) \\ &= \bar{\boldsymbol{\gamma}}^\top \boldsymbol{\Sigma} \bar{\boldsymbol{\gamma}} + \boldsymbol{\theta}^{*\top} \text{Var}(\bar{\boldsymbol{\gamma}}) \boldsymbol{\theta}^*. \end{aligned} \quad (41)$$

For an irreducible, aperiodic Markov chain with transition matrix $\boldsymbol{\Pi}$ and stationary distribution $\boldsymbol{\pi}^*$, the fundamental matrix [Kemeny and Snell, 1976] is $\mathbf{Z} = (\mathbf{I} - \boldsymbol{\Pi} + \mathbf{1}\boldsymbol{\pi}^{*\top})^{-1}$. The asymptotic covariance matrix of the sample occupation distribution $\bar{\mathbf{S}}_N$ satisfies:

$$\lim_{N \rightarrow \infty} N \text{Cov}(\bar{\mathbf{S}}_N) = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \boldsymbol{\pi}^* \boldsymbol{\pi}^{*\top}, \quad (42)$$

where $\mathbf{D}_\pi = \text{diag}(\boldsymbol{\pi}^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\boldsymbol{\Pi})$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (43)$$

Since $\theta_{\text{ATE}}(S) = \theta_0^* + (\theta_1^* - \theta_0^*) \bar{S}_N$, multiplying by $(\Delta\theta^*)^2 = (\theta_1^* - \theta_0^*)^2$ yields the total population variance $\sigma_{\text{total}}^2 = \sigma_{\text{cond}}^2 + \text{Var}(\bar{S}_N)(\Delta\theta^*)^2$. ■

F WHO 2021 Concentration-Response & Policy Valuation

Health impact calculations follow the official WHO 2021 Global Air Quality Guidelines systematic meta-analysis for all-cause non-accidental mortality. The attributable annual deaths avoided ΔM from an annual mean $\text{PM}_{2.5}$ reduction $\Delta \text{PM}_{2.5}$ is:

$$\Delta M = y_0 \cdot \text{Pop} \cdot (1 - \exp(-\beta \cdot \Delta \text{PM}_{2.5})), \quad (44)$$

where baseline mortality $y_0 = 7.2/1000$ per year (Registrar General of India SRS), and $\beta = \ln(1.062)/10.0 \approx 0.006015$ corresponds to pooled relative risk $\text{RR} = 1.062$ per $10 \mu\text{g}/\text{m}^3$. Economic evaluation applies a Value of Statistical Life (VSL) of \$0.45M USD per life saved, grounded in Indian empirical environmental health economics [Dominici et al., 2014].